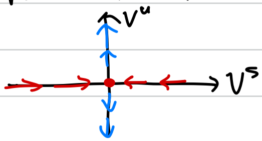
Reference Smale "Differentiable Dynamics"

Wiggins "Introduction to Applied Nonlinear Dyn. Systems and Chaos" Ch. 23~26

$f : M \longrightarrow M$ diffeo, $f(p) = p$.

p is hyperbolic if eigenvalues of d_pf satisfies $|\lambda| \neq 1$.

$\Rightarrow T_pM = V_p^s \oplus V_p^u$, where V^s is stable ($|\lambda|<1$) and V^u is unstable. ($|\lambda|>1$)



Note If f is sympl, $\dim V^s = \dim V^u$.

Stable manifold theorem

Let p be a hyp. f.p. Then $\exists$ contraction $g : W^s(p) \longrightarrow W^s(p)$ with $g(p_0) = p_0$ and injective immersion $J : W^s(p) \longrightarrow M$ s.t. $J(p_0) = p$,

$d_{p_0}J : T_{p_0}W^s(p) \longrightarrow T_pM$ is isom. onto V^s.

Moreover, $J(W^s(p)) = \{x \in M : f^m(x) \longrightarrow p \text{ as } m \to \infty\}$.

Call $W^s(p)$ or $J(W^s(p))$ (usually abused) stable manifold,

and $W^u(p)$, which is st. mfd. of f^{-1} unstable manifold.

Generic assumptions

(1) $\Omega = \{x \in M : \cup_{|m|>0} f^m(U) \cap U \neq \emptyset \ \forall x \in U\}$ non-wandering set.

Ω is finite, so $\Omega = \mathcal{P}(f)$

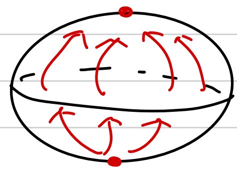
(2) $\forall$ Periodic points are hyperbolic.

(3) $\forall p, q \in \mathcal{P}(f)$, $W^s(p) \pitchfork W^u(q)$.

Ex $z \longrightarrow 2z$ on $\mathbb{CP}^1$. $\Rightarrow \mathcal{P}(f) = \{0, \infty\}$

$W^s(0) = \{0\}$, $W^u(0) = S^2 \setminus \{\infty\}$

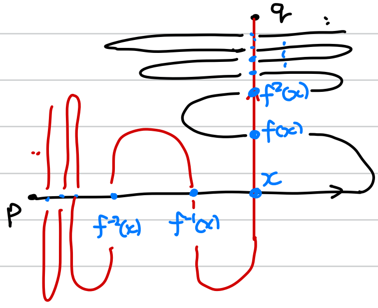
$W^s(\infty) = S^2 \setminus \{0\}$, $W^u(\infty) = \{\infty\}$



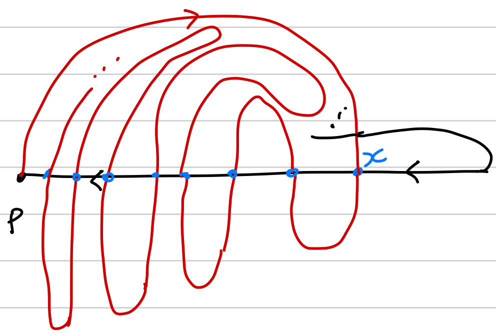
Consider $\dim M = 2$.

$x \in W^u(p) \cap W^s(q)$ is <u>heteroclinic</u> if $\dim W^u(p) = \dim W^s(q) = 1$

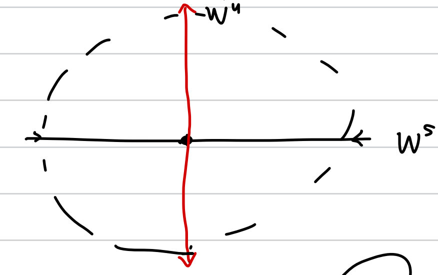
$\Rightarrow f^m(x) \in W^u(p) \cap W^s(q) \quad \forall m$.



$x \in W^u(p) \cap W^s(p)$ is called <u>homoclinic</u>.

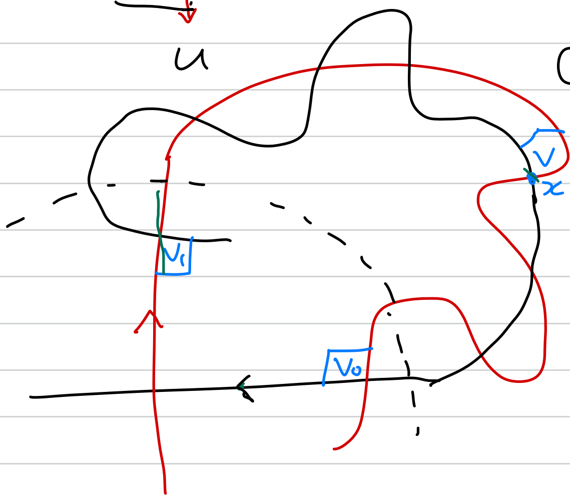


Let's look near p in the case of dim 2, $p=0$.

① 

U

② For homoclinic point x, we take k_0, k_1 large s.t. $f^{k_0}(x), f^{-k_1}(x) \in U$.

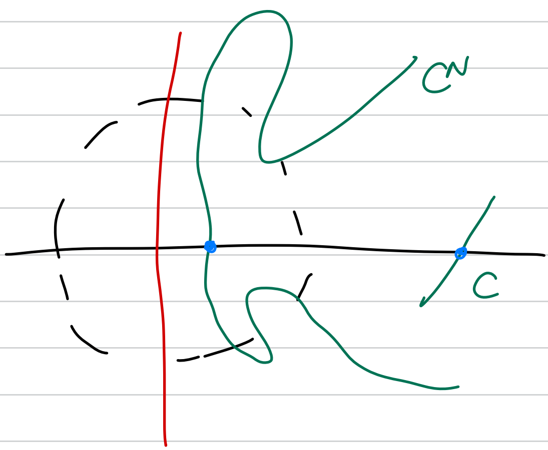


③ $f^{k_0}(V)=V_0$, $f^{-k_1}(V)=V_1$, $f^{k}(V_1)=V_0$ $(k=k_0+k_1)$

④ (λ-lemma)

Let C be a curve transversally intersecting $W^s(0)$ at $q \neq p$, C^N = Cpnt of $f^N(C) \cap U$ which contains q.

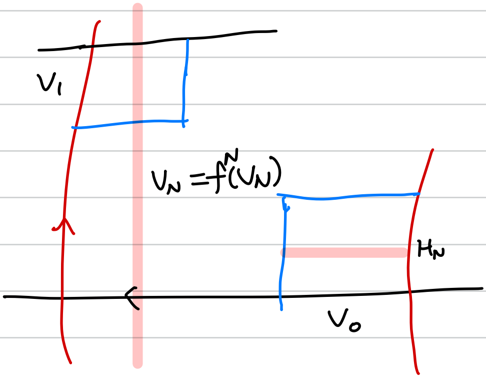
Then $\exists N$ large s.t. C^N is close to $W^u(0) \cap U$.



$f^k(V_1) = V_0$

Let $\mathrm{Dom}(f^T) = \{ p \in V_0 : \exists n \text{ s.t. } f^n(p) \in V_1 \text{ and } f(p), \cdots, f^{n-1}(p) \in V_0 \}$
and define $f^T(p) = f^n(p) \in V_1$, where n is smallest such n.

$\Rightarrow f^k \circ f^T : \mathrm{Dom}(f^T) \subset V_0 \longrightarrow V_0$ is defined.

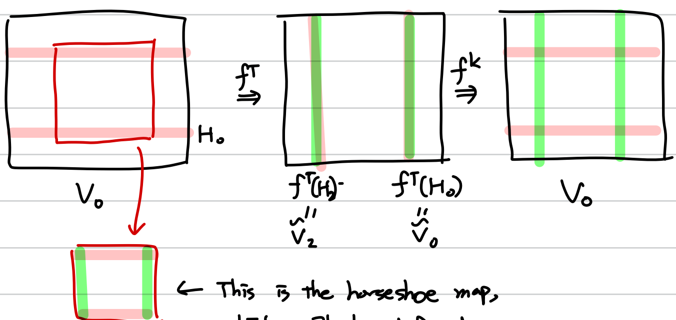


⑤ (Moser)
$F = f^k \circ f^T$ has invariant Cantor set, and F is top. conj. to a full shift on N symbols.

⌈Let $\tilde{V}_N = f^N(V_0) \cap V_1$
$f^{-N}(\tilde{V}_N) = H_N$.

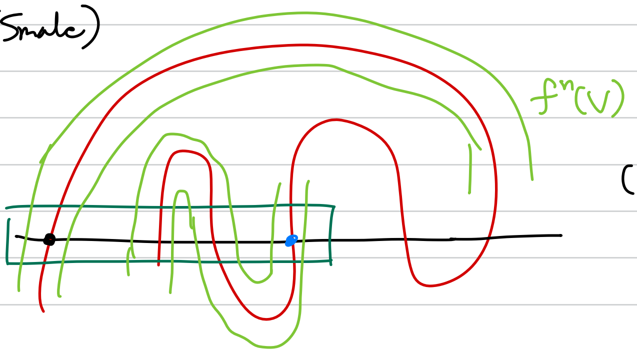
Choose $j_1 < j_2 < \cdots$ s.t. $\tilde{V}_i = f^{N_0 + j_i}(V_0) \cap V_1$ are disjointed, and define $f^{-N_0 - j_i}(\tilde{V}_i) = H_i$

$\Rightarrow$



← This is the horseshoe map, which will be defined soon.

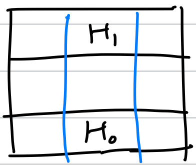
⑥ (Smale)

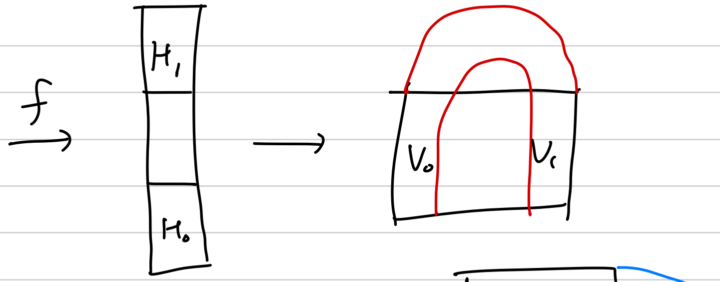


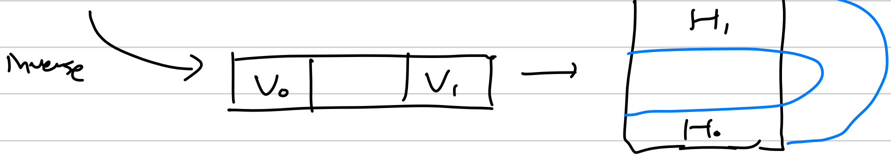
Taking appropriate nbhd V, $f^n(V)$ meets V finite times.

(return time is the same)

Horseshoe map







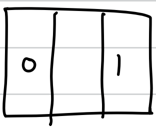
Want Invariant set $\bigcap_{n\in\mathbb{Z}} f^n(D)$

$\bigcap_{n\geq 0} f^n(D)$ is (Cantor set) $\times I$.

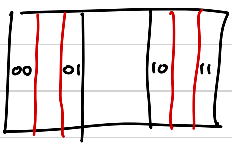
$\bigcap_{n\leq 0} f^n(D)$ is $I \times$ (Cantor set)

$\rightarrow$ labelled by biinfinite seq. $\cdots s_{-1}s_0s_1\cdots$ of 0 or 1.

Why?



$\Rightarrow$



$\Rightarrow \cdots$

We can use symbolic dynamics for this, with f = shift map.

Note Cantor set is characterized by closed perfect tot. disconnected set.

Note $\exists$ Countably many per. pts. of all possible period.
Also $\exists$ dense orbit.

Shift map

Goal Analyze the dynamics on the invariant set of horseshoe map.

Let $N = \{0, \cdots, N-1\}$, $S = \times_{\mathbb{Z}} N = \{(s_i) : i \in \mathbb{Z}\}$
$= \{\cdots s_{-2}s_{-1}.s_0 s_1 s_2 \cdots : s_i \in N\}$

Define $d(s, \bar{s}) = \sum \frac{1}{2^{|i|}} \frac{d_i(s_i, \bar{s}_i)}{1 + d_i(s_i, \bar{s}_i)}$, so S is a metric space,
where $d(i, j) = \delta_{ij}$.

Idea Nbhd U of (s_i) is $U = \{\bar{s} : s_i = \bar{s}_i \text{ for all } |i| \le M\}$
for some M.

Claim S is compact, tot. disconnected perfect set.

Thm (Brouwer)
Any two nonempty cpt Hdf spaces without isolated points with countable clopen basis are homeomorphic.
Let X be the space with basis $\mathcal{B} = \{B_1, B_2, \cdots\}$.
Want Define clopen sets U_s for each seq. $s \in 2^{\mathbb{N}}$.
① $U_\emptyset = X$
② If we defined $U_s \ne \emptyset$ for some $|s| = n$,
we can find $B_{n+1} \in \mathcal{B}$ s.t. $U_s \cap B_{n+1} \ne \emptyset$, $U_s \setminus B_{n+1} \ne \emptyset$. Why?
Define $U_{s0} = U_s \cap B_{n+1}$, $U_{s1} = U_s \setminus B_{n+1}$.
Time to review topology!

Define $f : X \longrightarrow C$ where $x \in \bigcap_{n=1}^{\infty} U_{s|n}$, then it's homeo!
$x \longmapsto (s_i)$
$\boxtimes$

Each point $p \in \Lambda = \bigcap f^n(D)$ is char. by (s_i) by $f^i(p) \in H_{s_i}$,
$f^{-i+1}(p) \in V_{s_{-i}}$.

$\Rightarrow f$ acts on Λ by shift map, $(s_i) \mapsto (s_{i+1})$. (or $f^{-i}(p) \in H_{s_{-i}}$)

Further topic?

1. Topological entropy. (In surface dynamics, horseshoe governs entropy.)
   「Katok 80」
2. More on symbolic dynamics?
3. Chaos?